

Name: _____

Exam 1 - MATH-262-02

Instructions: You have 110 minutes to take this exam. Please answer each question in the space provided. Answers without justification may **earn no credit**. You are permitted your signed prep work journal and a calculator on this exam. No other notes or electronic devices of any kind are allowed on this exam. Failure to comply may result in earning a 0 on any portion of this exam and/or being reported to the university.

Note: Unless otherwise stated, you may give implicit solutions.

1. Classify the differential equation

$$x^2 y'' + xy' + x^2 y = y + \ln(x)$$

according to its type, order, linearity, and homogeneity.

2. Verify that

$$r(\theta) = \frac{1}{\frac{1}{2} - \ln |\theta|}$$

is a solution to the equation

$$\frac{dr}{d\theta} = \frac{r^2}{\theta}$$

and find its largest interval of existence.

3. Determine whether the differential equation

$$6y^2 x \, dx - dy = 0$$

is separable, exact, and/or linear.

4. Solve

$$2\frac{du}{dt} - u = 4\sin(3t).$$

5. Solve

$$\frac{2ty}{t^2 + 1} - 2t - (2 - \ln(t^2 + 1))y'(t) = 0.$$

6. Use the substitution

$$u = 4x - y$$

to solve the IVP

$$y' = e^{4x-y}$$

While you may give your solution implicitly, it should not include u .

7. A 1000 gallon tank catches runoff from a chemical process. Suppose the tank initially contains 800 gallons of polluted water with 2 oz of pollutants. Polluted water runoff begins to flow into the tank at 3 gal/min. The runoff contains 5 oz of pollutants per gallon. The solution is piped to a water treatment plant at 3 gal/min.

(a) Write a first-order IVP that models the concentration (in oz/gal) of pollutants p after t minutes.

(b) Solve the equation you defined in part (a) using any method.

(c) How long does it take for the tank to accumulate 500 oz of pollutants?

(d) Suppose that regulation required the chemical process to be slowed once the tank accumulates 500 oz of pollutants in such a way that only 2 gal/min of runoff is produced. How would this affect the ODE model you wrote in part (a)? The algebraic model you found in part (b)?